

Certified Algorithms for Combinatorial Optimization Problems

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1 Introduction

The FEEDBACK VERTEX SET (FVS) problem is central to the field of computer science. Karp’s paper [1] showed that there is a reduction from SATISFIABILITY to twenty-one problems, one of which is FVS. Because of the hardness of the problem, extensive research has been conducted in order to find a good approximation solution. The problem is **APX-complete** and the best approximation ratio so far is two and was first found by Becker and Geiger [2]. Around the same time, Bafna et al. [3] devised the first local ratio technique in order to arrive at the same approximation ratio. These algorithms were translated in primal-dual terms by Chudak et al. [4] who constructed a third algorithm which yield a 2-approximation both for the unweighted and weighted case of the FVS problem.

For planar graphs and H-MINOR-FREE graphs two approaches have yielded PTASs for many problems. Baker’s technique [5] focused on approximation algorithms for **NP-Complete** problems on planar graphs, while Demaine and Hajiaghayi [6] generalized the shifting technique for the latter class of graphs. One of the known limitations of the shifting technique is that it does not handle non-local problems, such as the ones we are examining (FVS, CYCLE PACKING). In our paper, we impose restrictions on the problems in order to introduce a local property that allows us to devise a PTAS for the FVS-BCL problem. Baker’s technique allows for the translation into the weighted version of the problem. We adapt the algorithm to work for the MWFVS-BCL all the while trying to follow the framework for devising certified algorithms established in [7, 8, 9].

The FVS problem has applications in the field of Bayesian networks. As we said, researchers turned to approximation algorithms and made use of the linear programming (LP) framework in order to get desirable ratios. In this paper, we make use of the existing literature on LP for FVS and analyse how to translate the results for the class of graphs with bounded cycle length. In Section 2.3, we familiarize the reader with two ways of designing certified algorithms. One way is by using convex relations. This motivates us to define the problem as an ILP, solve the LP relaxation and exploit a randomized rounding scheme to get the values for the approximate and co-approximate conditions, as defined in the same section. Those values are then used to calculate the γ for the γ -certified algorithm.

Finally, we consider the notions of Bilu-Linial stability [10] and certified algorithms [7]. In the last two paragraphs, we explained how we plan to translate existing techniques to get certified solutions of instances. In the final work, the last sections will be dedicated to proving that these algorithms are indeed certified.

Related Work

This work is inspired by Baker’s seminal paper on designing PTASs in planar graphs [5]. There have been other techniques that generalize this approach, namely Demaine and Hajiaghayi [6]’s bidimensionality theory. The latter framework explores problems in H-MINOR-FREE graphs.

The notion of Bilu-Linial stability has given rise to much exploration of structured instances in many problems. In their paper, Bilu and Linial [10] offer the first evidence that stable instances are much easier than worst-case instances of problems. They propose an exact algorithm for $O(n)$ -stable instances of MAX CUT. This result was improved by Bilu et al. [11] who designed a $O(\sqrt{n})$ for the same problem. Further research yielded a general approach to analysing stable instances in Makarychev et al. [12]. They applied that approach to graph partitioning problems and designed $O(\sqrt{\log n \log \log n})$ -stable instances of MAX CUT and 4-stable instances of MINIMUM MULTIWAY CUT. The study of perturbation resilience of clustering problems is explored in [13, 14, 15]. On the TRAVELLING SALESMAN PROBLEM, Mihalák et al. [16] gave an algorithm for 1.8-stable instances. We refer the reader to the survey [17].

On the subject of Beyond the Worst-Case Analysis, we refer to Makarychev and Makarychev [7]. In that paper certified algorithms are first described in an attempt to bridge the gap between the worst-case and structured instances. Since then, researchers have made use of the definitions and results in that paper and applied them to other problems. Angelidakis et al. [8] explore the notion of stability and initiate the study

of certified algorithms for the MAXIMUM INDEPENDENT SET problem (MIS). They provide robust and certified algorithms for $(1+\epsilon)$ -stable instances of planar MIS, for any fixed $\epsilon > 0$. Covering and Dominating in planar graphs was further explored by Bumpus et al. [9]. They provide $(1 + \epsilon)$ -certified algorithms for DOMINATING SET and H-SUBGRAPH-FREE-DELETION, which for any ϵ run in time $f(1/\epsilon) \cdot n^{O(1)}$ on minor-closed classes of graphs of bounded local treewidth.

2 Preliminaries

In this section, we provide the reader with the necessary information in order to follow the paper. We start with a reminder of Graph Theory. After that, we move to more advanced definitions in the field of Combinatorial Optimization. Finally, we get on topic by introducing perturbation resilience and certified algorithms from the field of Beyond the Worst-Case Analysis of Algorithms.

2.1 Graph Theory

In this section, the authors refer to Bondy and Murty [18]. A *graph* G is an ordered pair denoted as $(V(G), E(G))$. The set $V(G)$ is called a set of *vertices* and the set $E(G)$ is a set of *edges*. The incidence function ψ_G associates each edge with an unordered pair of (not necessarily unique) vertices. The ordered pair and the incidence function define the graph G . We are going to use n as the cardinality of the set V and m for the set of edges E . With each edge e of G , let there be associated a real number $w(e)$, called its *weight*. Then G , together with the weighting function $w : E \rightarrow \mathbb{R}$, is called a *weighted graph*, and denoted (G, w) .

A graph can be *embedded in the plane* or is *planar* if it can be drawn on the plane in a way that edges intersect each other only at their ends. Such a drawing is called a *planar embedding* of the graph. A graph G is called *outerplanar* if it has a planar embedding $\tilde{G}$ in which all the vertices lie on the boundary of the outer face. This is also called a *1-outerplanar graph*. Two forbidden minors of that class are the K_4 and $K_{2,3}$ graphs. In the next subsection, we familiarize the reader with Baker's approach [5]. Here, we define a ℓ -*outerplanar* to be a graph that has a planar embedding in which the vertices belong to at most ℓ concentric layers. Every planar graph is ℓ -*outerplanar* for some ℓ . Given a graph G , we look for the ℓ -*outerplanar* embedding of G for which ℓ is minimal.

2.2 Combinatorial Optimization

In this section, we refer to the book on Combinatorial Optimization by Korte and Vygen [19] and Williamson and Shmoys [20]. Some definitions can be traced to the end of the book by Cygan et al. [21].

2.2.1 NP-Complete Problems: Definitions and Variants

We are going to be looking at a couple of problems, namely the FEEDBACK VERTEX SET, the CYCLE PACKING problem and the TRAVELLING SALESMAN PROBLEM. All of those problems belong to the class of **NP-complete** problems. The decision variant of the FVS problem can be defined as:

Definition 2.1: Feedback Vertex Set (FVS)

Input: A graph G and an integer ℓ .

Question: Does there exist a set X of at most ℓ vertices of G such that $G - X$ is a forest?

The weighted optimization version of the problem can be defined as:

Definition 2.2: Minimum Weight Feedback Vertex Set (MWFVS)

Instance:

- An undirected weighted graph (G, w) , where $G = (V, E)$ and $w : V \rightarrow \mathbb{R}$.

Task: Find a vertex set $X \subseteq V$ such that:

- $G \setminus X$ is acyclic (i.e., a forest)
- The sum of the weights of the vertices in X is minimized.

In some sense, the CYCLE PACKING problem is dual to FVS. We use the definition from the blue book of Cygan et al. [21].

Definition 2.3: Cycle Packing

Input: A graph G and an integer ℓ .

Question: Does there exist in G a family of ℓ pairwise vertex-disjoint cycles?

The other problem we inspect is an edge-optimization problem, namely the TRAVELLING SALESMAN PROBLEM (TSP).

Definition 2.4: Travelling Salesman Problem (TSP)

Instance:

- A set of cities $C = \{c_1, c_2, \dots, c_n\}$.
- A distance function $w : C \cdot C \rightarrow \mathbb{R}$ that gives the distance between each pair of cities.

Task: Find a permutation π of the cities such that:

- The total distance travelled is minimized, i.e., minimize

$$\sum_{i=1}^{n-1} w(c_{\pi(i)}, c_{\pi(i+1)}) + w(c_{\pi(n)}, c_{\pi(1)}).$$

We define the problems FVS-BCL, MWFVS-BCL and CYCLE PACKING-BCL to be the problems that cover and pack cycles of bounded length. The 'BCL' stands for bounded cycle length. We give the following definitions:

Definition 2.5: FVS-BCL

Input: A graph G , integer ℓ and an integer r .

Question: Does there exist a set X of at most ℓ vertices such that every cycle of length at most r contains a vertex in X ?

The following is the weighted version of the problem; we define it as an optimization task.

Definition 2.6: MWFVS-BCL

Instance:

- An undirected weighted graph (G, w) , where $G = (V, E)$ and $w : V \rightarrow \mathbb{R}$ and an integer r .

Task: Find a vertex set $X \subseteq V$ such that:

- Every cycle in $G \setminus X$ of length $\leq r$ has a vertex in X .
- The sum of the weights of the vertices in X is minimized.

Finally, we define the maximization problem with an additional constraint:

Definition 2.7: Cycle Packing-BCL

Input: A graph G , an integer ℓ and an integer r .

Question: Does there exist in G a family of ℓ pairwise vertex-disjoint cycles with each of them having length $\leq r$?

2.2.2 Approximation Algorithms and Complexity Classes

As we mentioned, all the problems are **NP-complete**, so we turn to *approximation algorithms* in order to cope with the hardness of the problems. Let Π be an optimization problem with nonnegative weights and $k \geq 1$. A *k-factor approximation algorithm* for Π is a polynomial-time algorithm A for Π , such that

$$\frac{1}{k} \cdot \text{OPT}(I) \leq A(I) \leq k \cdot \text{OPT}(I)$$

for all instances I of Π . The first inequality shown is true for maximization problems, while the second one holds for minimization problems. We say A has a performance ratio of k . If a problem falls into that category, we say that it is **APX-complete**.

In the last paragraph, we mentioned the class of problems **APX** regarding approximation algorithms. Now we turn to another class of approximation algorithms. **PTAS** is a class of algorithms that work in polynomial time and get close to an optimal solution with a trade-off.

Definition 2.8: Polynomial-Time Approximation Scheme (PTAS)

A polynomial-time approximation scheme (**PTAS**) is a family of algorithms A_ϵ where there is an algorithm for each $\epsilon > 0$, such that A_ϵ is a $(1 + \epsilon)$ -approximation algorithm (for minimization problems). The running time of the algorithm A_ϵ is polynomial, but can depend arbitrarily on $1/\epsilon$. This dependence could be exponential or even worse.

Every problem with a **PTAS** is also in **APX**. The relation between these two classes is similar to the one between **P** and **NP**. For more on the topic, such as *APX-completeness*, *L-reductions* and the *PCP-theorem*, we refer to Korte and Vygen [19].

2.2.3 Baker's approach

In this subsection, we are going to describe the Shifting technique. It is described in Baker's seminal paper [5]. What she describes is a general algorithm that partitions the graph into a disjoint union of levels of vertices. By analysing slices of these levels and, more specifically, the overlap between these slices, one obtains an approximation ratio for the solution to the optimization problem. The slices make up several levels that are specifically chosen for a problem and it should hold that an exact solution can be found on one slice. It must also hold that this solution can be found in optimal time.

There is also another way to view this approach. Instead of analysing the overlap between slices, one could partition the graph by removing said overlap. This way we get a disjoint union which does not sum up to the total solution, but we can argue that we did not stray away so much from the optimum solution. This approach works for hereditary maximization problems like INDEPENDENT SET, MAX-CUT, MAXIMUM P-MATCHING, etc.

We are going to give two examples. The first is the general shifting technique and the example problem for it will be VERTEX COVER. Then we will explore the second perspective and take the INDEPENDENT SET problem as an example. It is important to note that the general technique is much more powerful than partitioning by removal of vertices. With VERTEX COVER, we can still use the second approach and get a

PTAS, but the same does not hold for the DOMINATING SET problem. However, using the general shifting technique, we can obtain a PTAS for DOMINATING SET in planar graphs. This was done by Marzban and Gu [22]. They designed a PTAS with an approximation ratio of $(1 + 2/\ell)$ in $O(2^\ell \cdot n)$ time. The two in the approximation ratio appears because of the overlap needed for DOMINATING SET between slices, which is two levels.

What follows is a brief explanation of Baker's shifting technique with VERTEX COVER as an example.

Algorithm 1 Baker's shifting technique

Require: Planar graph $G = (V, E)$, parameter ℓ

Ensure: $(1 + \epsilon)$ -approximation for VERTEX COVER

- 1: Perform BFS from some arbitrary vertex r
 - 2: $S \leftarrow \emptyset$
 i : *shift*; j : *slice*
 - 3: **for** each $i = 0$ to $\ell - 1$ **do**
 - 4: Let $G_{i,j}$ be the subgraph induced on vertices at levels $j \cdot \ell + i$ through $(j + 1) \cdot \ell + i$ for $0 \leq i < \ell$.
 - 5: $\forall j$ compute $S_{i,j}$, the min. VC of $G_{i,j}$.
 - 6: $\forall j$ let $S_i = \bigcup_j S_{i,j}$
 - 7: $S \leftarrow S \cup S_i$
 - 8: **end for**
 - 9: **return** S_{i^*} from S with minimum cardinality
-

Analysis of the algorithm:

Let V_i be the set of vertices at levels $i \bmod \ell$:

$$V_i = \{v \in V \mid \text{level}(v) \equiv i \bmod \ell\}$$

Let OPT_i denote $OPT \cap V_i$. We notice that OPT_i are disjoint and $OPT = \bigcup_i OPT_i$. Therefore, we can make the following argument:

1. $\exists q : |OPT_q| \leq \frac{1}{\ell} \cdot OPT = \epsilon |OPT|$
2. $S_q \leq \sum_j S_{q,j}$, for some shift q we have this inequality because in step 5. of the algorithm we define the set to be a union over all slices and some vertices may appear twice in the boundary layers.
3. $\sum_j S_{q,j} \leq \sum_j |OPT \cap V(G_{q,j})|$, we know that $S_{q,j}$ is a min. VC for $G_{q,j}$, while the optimal solution may choose different vertices, especially on the boundary for the global optimality of the solution.
4. $\sum_j |OPT \cap V(G_{q,j})| \leq |OPT| + |OPT_q|$, if we consider the graphs $\{G_{q,j}\}$, then we can argue that each vertex appears once, except those at levels $q \bmod \ell$, which appear twice.

From the first, second and fourth step we can conclude that:

$$S_q \leq |OPT| + |OPT_q| \leq (1 + \epsilon) \cdot |OPT|$$

Now we take a look at the second approach we described.

Algorithm 2 Baker's technique with removal

Require: Planar graph $G = (V, E)$, parameter ℓ

Ensure: $(1 - \epsilon)$ -approximation for MIS

```
1:  $S \leftarrow \emptyset$ 
2: Perform BFS from some arbitrary vertex  $r$ 
3:  $\ell \leftarrow 1/\epsilon$ 
4: Label each vertex according to its layer  $i \bmod \ell$ :  $V_1, V_2, \dots, V_{\ell-1}$ 
5: for each  $i = 0$  to  $\ell - 1$  do
6:   Let  $G_i \leftarrow G \setminus V_i$ 
7:   Let  $G_1, G_2, \dots, G_j$  be the connected components of  $G_i$ 
8:    $\forall j$  compute the MIS  $S_{i,j}$  of  $G_{i,j}$ 
9:    $S_i \leftarrow \bigcup_j S_{i,j}$ 
10:   $S \leftarrow S \cup S_i$ 
11: end for
12: return  $S_{i^*}$  from  $S$  with minimum cardinality
```

Consequence of the algorithm:

1. for $\ell = 1/\epsilon$ compute partition $P_1, \dots, P_k$.
2. for some q ($1 \leq q \leq \ell$), $|OPT \cap P_q| \leq \frac{1}{\ell} \cdot |OPT| = \epsilon \cdot |OPT|$ because of the pigeonhole principle
3. hence, $|OPT \cap (G - P_q)| \geq (1 - 1/\ell) \cdot |OPT| = (1 - \epsilon) \cdot |OPT|$
4. return $\max_i |\text{MIS}(G - P_i)|$

We skip much of the analysis that we did for the shifting technique, however, we argue that for this particular problem for some P_q , a small fraction of the solution resides within that partition. Therefore, we can forget about that part and the rest of the solution will still be an independent set of bounded size.

The first approach is more general and powerful for constructing algorithms, but it doesn't apply well to the DOMINATING SET problem. The key issue is that the removal technique relies on intersecting an optimal solution with a subgraph, expecting it to remain valid for the subgraph. This fails for DOMINATING SET, since a vertex in the subgraph could still be dominated by a vertex outside the subgraph.

It is shown that the general shifting technique can be used to obtain a PTAS for DOMINATING SET. The interesting part is that in order to properly analyse the solution, one must make the slices overlap on two levels. We refer to Marzban and Gu [22] for the explanation as to why the last is true and how to obtain the $(1 + 2/\ell)$ approximation ratio.

2.2.4 Linear Programming and Randomized Rounding

One way of approximating solutions is done through Linear Programming (LP). For a more detailed review of LP, the reader can consult Chapter 3 of Korte and Vygen [19]. The algorithm we use solves a linear relaxation, which provides fractional values as solutions. We then round this fractional solution to an integral one, but rather than doing it deterministically, we do the process randomly. This technique is called *randomized rounding*. We exploit that framework, because it is one of the ways one could devise a certified algorithm as explained in Makarychev and Makarychev [7].

2.3 Beyond the Worst-Case Analysis of Algorithms

In this section we explore perturbation-resilience, also known as γ -stability or Bilu-Linial stability. The authors refer to the original paper of the two authors Bilu and Linial [10] as well as a survey on the matter

by Makarychev and Makarychev [17]. In a later work, Makarychev and Makarychev define γ -certified algorithms [7]. In the final paragraph, we take a look back at combinatorial optimization problems through the lens of constraint satisfaction.

The following definitions for γ -perturbation are for edge-optimization and vertex-optimization problems, respectively. Afterwards, we are going to define stability and certified algorithms for vertex-optimization problems only.

Definition 2.9: γ -perturbation for Edge-Optimization problems

Consider a weighted graph $G = (V, E, w)$ with a set of edge weights w_e . An instance $G' = (V, E, w')$ is a γ -perturbation ($\gamma \in \mathbb{R}_{\geq 1}$) of G if $w(e) \leq w'(e) \leq \gamma \cdot w(e)$; that is, if we can obtain a perturbed instance from the original by multiplying the weight of each edge by a number between 1 and γ (the number may vary for each edge).

From now on, the definitions will concentrate on Vertex-Optimization problems.

Definition 2.10: γ -perturbation for Vertex-Optimization problems

Consider a weighted graph (G, w) . For any $\gamma \in \mathbb{R}_{\geq 1}$ a γ -perturbation of the weight function $w : V \rightarrow \mathbb{R}$ is a function $w' : V \rightarrow \mathbb{R}$ satisfying $w(v) \leq w'(v) \leq \gamma \cdot w(v) \forall v \in V$.

Definition 2.11: γ -stability

For any $\gamma \in \mathbb{R}_{\geq 1}$, a γ -stable instance (G, w) of a *vertex-minimization* problem Π is an instance admitting a unique optimal solution S which remains optimal even under γ -perturbations of (G, w) .

Definition 2.12: Certified Algorithm

A γ -certified solution to an instance (G, w) of a weighted vertex-optimization problem Π is a pair (S, w') where w' is a γ -perturbation of w and S is an optimal solution on (G, w') . A γ -certified algorithm for Π maps instances of Π to γ -certified solutions.

Definition 2.13: Combinatorial Optimization problems (Constraint Satisfaction)

An instance of a *combinatorial optimization* problem is specified by a set of feasible solutions $\mathcal{S}$ (the solution space) a set of constraints $\mathcal{C}$ and constraint weights $w_c > 0$ for $c \in \mathcal{C}$. Typically the solution space is exponential in size (as is the case with MWFVS with the number of cycles in a graph) and is not provided explicitly. We say that a feasible solution $s \in \mathcal{S}$ satisfies a constraint $c \in \mathcal{C}$ if $c(s) = 1$.

Consider the minimization and maximization objectives.

- The minimization objective is to minimize the total weight of the unsatisfied constraints: find $s \in \mathcal{S}$ that minimizes $\sum_{c \in \mathcal{C}} w_c \cdot (1 - c(s)) = w(\mathcal{C}) - \text{val}_{\mathcal{I}}(s)$, where $w(\mathcal{C}) = \sum_{c \in \mathcal{C}} w(c)$ is the total weight of the constraints.
- The maximization objective is to maximize the total weight of the satisfied constraints: find $s \in \mathcal{S}$ that maximizes $\text{val}_{\mathcal{I}}(s) = \sum_{c \in \mathcal{C}} w(c) \cdot c(s)$.

Now we familiarize the reader with the two ways of designing certified algorithms as presented in [7]. The first is a general framework that (1) starts with an arbitrary solution and (2) iteratively improves it. The improvements in step 2, however, are not necessarily local, though it may look like a local search technique.

The following is taken from the paper on certified algorithms where it is presented as Task 5.13.

Task 2.14: General framework for γ -certified algorithms

Assume we are given an instance $\mathcal{I}(\mathcal{S}, \mathcal{C}, w)$, a partition of its constraints $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2$ and a parameter γ . The task is to either:

- Option 1: find a better $s \in \mathcal{S}$, such that $\gamma \cdot \sum_{c \in \mathcal{C}_1} w_c c(s) > \sum_{c \in \mathcal{C}_2} w_c (1 - c(s))$, or
- Option 2: to report that for every $s \in \mathcal{S}$: $\sum_{c \in \mathcal{C}_1} w_c c(s) \leq \sum_{c \in \mathcal{C}_2} w_c (1 - c(s))$.

In this procedure, $\mathcal{C}_1$ and $\mathcal{C}_2$ are the sets of constraints that are currently unsatisfied and satisfied, respectively. If $\gamma = 1$, then Option 1 would try to find a solution where the weight of the unsatisfied constraints is strictly greater than the weight of the unsatisfied by s . Therefore, Option 1 finds a better solution than the current one, while Option 2 reports that there is no solution better than s . The options are not mutually exclusive.

For the second approach, we refer the reader to the paper of Makarychev and Makarychev [7] as the explanation is quite long and needs to be understood with the examples given.

3 Research Question

In this work, we explore several questions. The first two are of most significance, while the problems that follow will be tackled in what time is left.

- Explore the FVS-BCL problem as defined in Definition 2.5 for planar graphs.
 - To that end we are going to make use of Baker’s shifting technique [5] as presented in Algorithm 1.
 - By using the modified version with an overlap between layers of $\lfloor \frac{r}{2} \rfloor + 1$, we capture the essence of breaking cycles of bounded length r .
 - Baker’s technique is known to work for problems that involve local conditions on nodes and edges and our imposed restrictions on the FVS allow us to deploy this framework.
 - We extend this method to devise a PTAS for the weighted version of the problem, MWFVS-BCL, as defined in Definition 2.6, for planar graphs.
- Initiate the study of certified algorithms for MWFVS-BCL. We provide robust and certified algorithms for $(1 + \epsilon)$ -stable instances of planar MWFVS-BCL, for any fixed $\epsilon > 0$.
- Future Directions:
 - Investigating the CYCLE PACKING-BCL problem. As a hereditary maximization task, we make use of the alternative approach to Baker’s technique, namely Algorithm 2 with removal.
 - Exploring the translation of PTASs for the TSP to $(1 + \epsilon)$ -certified algorithms.

4 (E)PTAS for the FVS-BCL Problem Using Baker's Technique

In this section we will demonstrate how to obtain an EPTAS for the weighted version of FVS-BCL. We begin by looking at the unweighted version and argue about the correctness and efficiency of our algorithm. Then we prove that using tree decompositions and dynamic programming one can exactly solve the 4-cycle-breaking problem in one subgraph. Finally, we show that the running time is indeed polynomial in the size of the input (and exponential in the parameter ℓ which we define below).

Before we start, note that in subsection 4.1 and subsection 4.2 the algorithms that are provided do not have an input variable for the length of the cycle. As we said, we assume that this variable ρ is implicitly set to four, i.e. our algorithm is only concerned with breaking 4-cycles. In subsection 4.3 we analyse the algorithm for an arbitrary value for ρ .

4.1 Baker's technique for unweighted FVS-BCL on planar graphs

In this subsection we provide the necessary definitions, lemmas and theorems to demonstrate how to obtain an EPTAS for the problem we defined in 2.5, namely FVS-BCL. In the preliminaries we gave two examples of how to construct algorithms using Baker's technique and how to further analyse them. In the same fashion we now construct Algorithm 3 and reason for its effectiveness.

In the following algorithm we use the letter ℓ for the outerplanarity of each subgraph, i.e. each $G_{i,j}$ that we define on line 4 in Algorithm 3 is ℓ -outerplanar.

Algorithm 3 Baker's technique for the unweighted FVS-BCL

Require: Planar graph $G = (V, E)$, parameter ℓ

Ensure: $(1 + \epsilon)$ -approximation for FVS-BCL

- 1: Perform BFS from some arbitrary vertex r
 - 2: $S \leftarrow \emptyset$
 i : *shift*; j : *slice*
 - 3: **for** each $i = 0$ to $\ell - 1$ **do**
 - 4: Let $G_{i,j}$ be the subgraph induced on vertices at levels $j \cdot \ell + i$ through $(j + 1) \cdot \ell + i + 1$ for $0 \leq i < \ell$.
 - 5: $\forall j$ compute $S_{i,j}$, the min. unweighted FVS-BCL of $G_{i,j}$ using Algorithm 4 as a subroutine.
 - 6: $\forall j$ let $S_i = \bigcup_j S_{i,j}$
 - 7: $S \leftarrow S \cup S_i$
 - 8: **end for**
 - 9: **return** S_{i*} from S with minimum cardinality
-

In the analysis of the algorithm we are going to need three lemmas in order to prove that Algorithm 3 obtains a $(1 + 2/\ell)$ -approximation of the optimum solution.

Lemma 4.1: Optimum solution bound (Unweighted Case)

Let $S_{i,j}$ be an optimum solution for $G_{i,j}$ for some shift i and slice j , let $F \equiv OPT$ be the optimum solution in G and let $F_{i,j} = G_{i,j} \cap F$. In the unweighted case we claim that:

$$|S_{i,j}| \leq |F_{i,j}|.$$

We will say that $F_{i,j}$ is a FVS-BCL of $G_{i,j}$, not necessarily the one of minimum cardinality.

For now we do not argue about how we obtain an optimum solution, in further sections we will show how

to get it using tree decompositions and dynamic programming, but for now we argue that this inequality holds in each subgraph $G_{i,j}$.

We first argue that in each subgraph $G_{i,j}$, the set $F_{i,j}$ contains a vertex that breaks every 4-cycle. Let us fix a slice and shift a and b and take a look at the subgraph $G_{a,b}$, the optimal solution for it $S_{a,b}$ and the set $F_{a,b}$. Notice that, much like in the Vertex Cover analysis, no vertex in $V(G) \setminus V(G_{a,b})$ can break any 4-cycle that is contained within $G_{a,b}$. Therefore $F_{a,b}$ must contain a vertex that hits every 4-cycle or it will not be a solution in $G_{a,b}$ and subsequently in G . From this and the fact that $S_{a,b}$ is the optimum solution for $G_{a,b}$ we conclude that:

$$|S_{a,b}| \leq |F_{a,b}|.$$

for some (a,b) , therefore it holds $\forall(i,j)$.

One must confront the fact that notation is abused throughout the paper, especially when it comes down to j , the slice variable. When we say that a statement holds for all values i and j , we mean that it holds for the subgraphs that these variables outline. We will sketch an example to deepen our understanding of Lemma 4.1. Moreover, we are going to explore one avenue that gives rise to an inequality in the lemma.

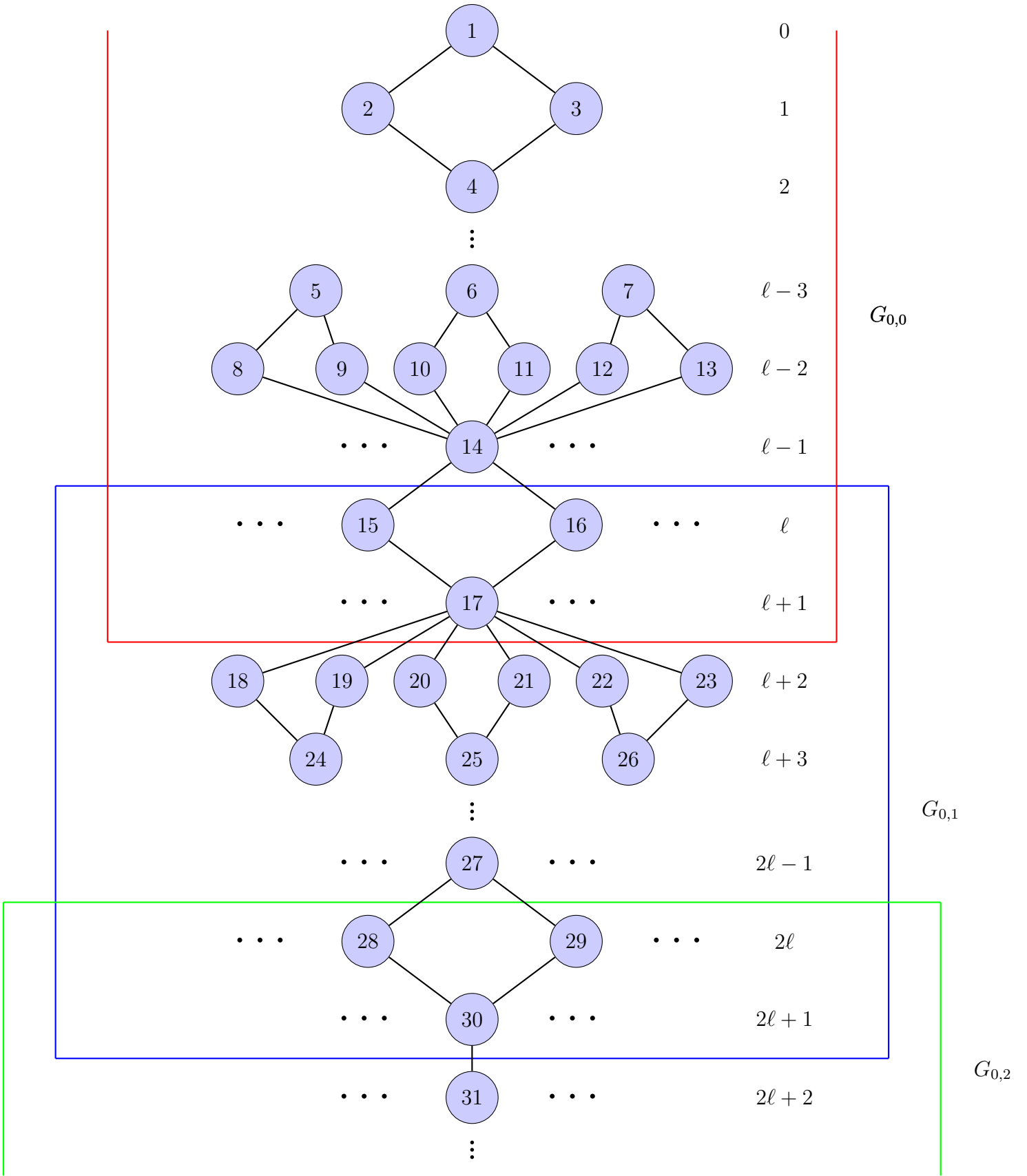


Figure 1: BFS tree, showing the distance from the root for each layer.

Let G look like the graph we sketched on the previous page. The vertical dots help us clarify the levels

while the horizontal signify that there may be other vertices of interest. However, there are vertices we would like to inspect, specifically vertices with labels 14 and 17. Let $v_{14}, v_{17} \in F$. What follows is that $v_{14}, v_{17} \in F_{0,0}$. The cycles that give us reason to claim that $v_{14} \in F$ are all within $G_{0,0}$. Therefore, it holds that Algorithm 3 will have $v_{14} \in S_{0,0}$. However, the same does not hold for v_{17} , the cycles that it is a part of are in $G_{0,1}$ and thus $v_{17} \notin S_{0,0}$, while $v_{17} \in F_{0,0}$. This example should not distract the reader from the fact the real reason that $|S_{0,0}| \leq |F_{0,0}|$ is because $S_{0,0}$ is by definition the minimum 4-cycle-breaking set for $G_{0,0}$, while $F_{0,0}$ is, as we previously argued, a feasible 4-cycle-breaking set.

What follows from Lemma 4.1 is a result that holds for any shift i .

Result 4.2: Optimum solution bound for the whole graph (Unweighted Case)

Let S_i be the union of optimal solutions defined on line 6 of Algorithm 3 for some shift i . Let $F \equiv OPT$ be the optimum solution in G and let $F_{i,j} = G_{i,j} \cap F$. For those sets it holds that:

$$|S_i| \leq \sum_j |S_{i,j}| \leq \sum_j |F_{i,j}|.$$

The first inequality in Result 4.2 stems from the fact that some vertices in the $S_{i,j}$ sum may appear twice. This pertains to vertices that are located on the boundary. For example if the $v_{30} \in S_{0,1} \wedge v_{30} \in S_{0,2}$, then it would be counted once in S_0 , but twice in $\sum_j S_{0,j}$. The second inequality is a result from Lemma 4.1 where we argue that $S_{i,j}$ is an optimum solution for $G_{i,j}$ while $F_{i,j}$ is a 4-cycle-breaking set, but not one of necessarily minimum cardinality.

In the Lemma 4.3 we look at the bound of the union of the $F_{i,j}$ sets. Before that, however, we make an argument for the vertices on the boundaries.

Lemma 4.3: Bound for the vertices on the boundaries (Unweighted Case)

Let $OPT_i = OPT \cap \{\text{vertices at levels } i \bmod \ell\}$. The sets OPT_i are disjoint and $\bigcup_i OPT_i = OPT$. We claim that:

$$\exists q \in \{0, 1, \dots, \ell - 1\} : |OPT_q| + |OPT_{q+1}| \leq \frac{2}{\ell} \cdot |OPT|.$$

Suppose that $\forall q : |OPT_q| + |OPT_{q+1}| > \frac{2}{\ell} \cdot |OPT|$.

As we said those sets are disjoint, meaning $OPT_q \cap OPT_{q+1} = \emptyset$. We make the following arguments:

1. If inequality ?? holds, then $\sum_{i=0}^{\ell-1} |OPT_i| + |OPT_{i+1}| > 2 \cdot |OPT|$.
2. Because each OPT_i exists twice in that sum and $OPT_i \cap OPT_j = \emptyset, \forall (i, j), i \neq j$, then the property $\bigcup_i OPT_i = OPT$ from which follows $\sum_i |OPT_i| = |OPT|$ would not hold.
3. From the contradiction in step two, we argue that there indeed $\exists q : q \in \{0, 1, \dots, \ell - 1\}$ by the *pigeonhole principle*, such that $|OPT_q| + |OPT_{q+1}| \leq \frac{2}{\ell} \cdot |OPT|$.

Now we argue for the size of the sets $F_{i,j}$. It is natural that we should be able to tie their size to the size of the optimal solution F . We will use Lemma 4.3 to do so. First we will state the lemma.

Lemma 4.4: Bound for the cardinality of the intersection sets (Unweighted Case)

Let $F \equiv OPT$ be the optimal solution and $F_{i,j} = F \cap G_{i,j}$ be the intersection sets with the subgraphs. Then we have:

$$\sum_j |F_{i,j}| \leq |F| + \frac{2}{\ell} \cdot |F|$$

for some specific integer i .

If we look at Fig. 1, then we notice that any vertex that is located in the overlap layers would be counted twice in the sum $\sum_j |F_{i,j}|$. The interior vertices are counted once towards the optimum solution sum.

Therefore, we have that:

1. $\sum_j |F_{i,j}| = |F| + |\{\text{vertices on the boundary}\}| = |F| + |OPT_i| + |OPT_{i+1}|$ for some $i \in \{0, 1, \dots, \ell-1\}$.
2. For the RHS of the equation in 1. we argue that there exists i^* , such that

$$|OPT_{i^*}| + |OPT_{i^*+1}| \leq \frac{2}{\ell} \cdot |OPT|$$

using Lemma 4.3.

3. For that i^* it holds that $\sum_j |F_{i^*,j}| \leq |F| + \frac{2}{\ell} |F|$.

Finally, using Lemma 4.1, Result 4.2 and Lemma 4.4 we chain the inequalities that are presented and get that for some slice i^* we have:

$$|S_{i^*}| \leq \sum_j |S_{i^*,j}| \leq \sum_j |F_{i^*,j}| \leq |F| + \frac{2}{\ell} \cdot |F|$$

Algorithm 4 Exact algorithm for unweighted FVS-BCL for each subgraph $G_{i,j}$

Require: ℓ -outerplanar graph $G = (V, E)$

Ensure: Exact solution S

1: **return** S

4.2 Baker's technique for the weighted FVS-BCL on planar graphs

In this subsection we look at the weighted version of the problem. Firstly, we construct Algorithm 5 which differs from Algorithm 3 only on step 5. For step 5 we construct Algorithm 6 and use it as a subroutine to get the solution we want.

Algorithm 5 Baker's technique for the weighted FVS-BCL

Require: Planar graph $G = (V, E)$, parameter ℓ , $w : V \rightarrow \mathbb{R}$

Ensure: $(1 + \epsilon)$ -approximation for FVS-BCL

- 1: Perform BFS from some arbitrary vertex r
 - 2: $S \leftarrow \emptyset$
 i: shift; j: slice
 - 3: **for** each $i = 0$ to $\ell - 1$ **do**
 - 4: Let $G_{i,j}$ be the subgraph induced on vertices at levels $j \cdot \ell + i$ through $(j + 1) \cdot \ell + i + 1$ for $0 \leq i < \ell$.
 - 5: $\forall j$ compute $S_{i,j}$, the min. weighted FVS-BCL of $G_{i,j}$ using Algorithm 6 as a subroutine.
 - 6: $\forall j$ let $S_i = \bigcup_j S_{i,j}$
 - 7: $S \leftarrow S \cup S_i$
 - 8: **end for**
 - 9: **return** S_{i^*} from S , such that $w(S_{i^*}) \leq w(S_i), \forall i \in \{1, 2, \dots, \ell - 1\}$.
-

Much like the proof we did in subsection 4.1, we start off with finding a bound for the optimum solution for each subgraph.

Lemma 4.5: Optimum solution bound (Weighted Case)

Let $G = (V, E)$ be a planar graph and $w : V \rightarrow \mathbb{R}$ be a weight function. Let $S_{i,j}$ be an optimum solution for $G_{i,j}$ for some shift i and slice j , let $F \equiv OPT$ be the optimum solution in G and let $F_{i,j} = G_{i,j} \cap F$. In the weighted case we claim that:

$$w(S_{i,j}) \leq w(F_{i,j}).$$

The proof of Lemma 4.5 is presented in three steps. Let us fix the integer arguments for shift and slice. We take a to be the offset (shift) and look at an arbitrary slice b . Note that when we look at a slice b , we mean that we look at the subgraph $G_{a,b}$ induced on vertices at levels $b \cdot \ell + a$ through $(b + 1) \cdot \ell + a + 1$.

1. Firstly, one needs to be sure that no vertex v , such that $v \in V(G) \setminus V(G_{a,b})$ can break any 4-cycle that is contained within $G_{a,b}$.
2. From step 1 and the definition of $F_{a,b}$, it follows that $F_{a,b}$ is a feasible solution of the 4-cycle-breaking problem within $G_{a,b}$.
3. Finally we argue, that because $S_{a,b}$ is an optimum solution for the 4-cycle-breaking problem within $G_{a,b}$ and $F_{a,b}$ is a feasible solution for the same problem, we have:

$$w(S_{a,b}) \leq w(F_{a,b})$$

The statement in Lemma 4.5 holds for all values for the shift and slice. Now we mention a useful result that follows from the lemma.

Result 4.6: Optimum solution bound for the whole graph (Weighted Case)

Let S_i be the union of optimal solutions defined on line 6 of Algorithm 5 for some shift i . Let $F \equiv OPT$ be the optimum solution in G and let $F_{i,j} = G_{i,j} \cap F$. For those sets it holds that:

$$w(S_i) \leq \sum_j w(S_{i,j}) \leq \sum_j w(F_{i,j}).$$

Proof. This result immediately follows from the fact that when we define S_i as the union of the subgraphs some vertices, specifically those on the overlapping layers, will have their weights added twice to the sum $\sum_j w(S_{i,j})$. If we inspect the LHS, however, each vertex and its weight will be counted only once towards $w(S_i)$. Therefore, it holds that $w(S_i) \leq \sum_j w(S_{i,j})$ and from Lemma 4.5 we chain the inequalities to get Result 4.6.

Lemma 4.7: Bound for the weight of the vertices on the boundary (Weighted Case)

Let $G = (V, E)$ be a planar graph and suppose we run a BFS from an arbitrary vertex. Let $w : V \rightarrow \mathbb{R}$ be a weight function. Let $OPT_i = OPT \cap \{\text{vertices at levels } i \bmod \ell\}$. The sets OPT_i are disjoint and $\bigcup_i OPT_i = OPT$.

We claim that:

$$\exists q \in \{0, 1, \dots, \ell - 1\} : w(OPT_q) + w(OPT_{q+1}) \leq \frac{2}{\ell} \cdot w(OPT).$$

Proof. Suppose the opposite, $\forall i \in \{1, 2, \dots, \ell - 1\}$ it holds that $w(OPT_i) + w(OPT_{i+1}) > \frac{2}{\ell} \cdot w(OPT)$.

1. From the two properties in the definition of OPT_i , namely that

$$OPT_i \cap OPT_j = \emptyset, \forall i, j \in \{1, 2, \dots, \ell - 1\}, i \neq j \text{ and}$$

$$OPT = \bigcup_i OPT_i, \text{ it follows that } \sum_{i=0}^{\ell-1} w(OPT_i) = w(OPT).$$

2. However, if inequality ?? holds, then $\sum_{i=0}^{\ell-1} w(OPT_i) + w(OPT_{i+1}) > \sum_{i=0}^{\ell-1} \frac{2}{\ell} \cdot w(OPT) > 2 \cdot w(OPT)$. Because each OPT_i is counted twice in the sum of step 2, then there is a contradiction between the sums in steps 1 and 2.
3. Finally, we claim that because of the *pigeonhole principle* there must exist a "small pair" OPT_q and OPT_{q+1} , such that $w(OPT_q) + w(OPT_{q+1}) \leq \frac{2}{\ell} \cdot w(OPT)$.

Lemma 4.8: Bound for the weight of the intersection sets (Weighted Case)

Let $F \equiv OPT$ be the optimal solution and $F_{i,j} = F \cap G_{i,j}$ be the intersection sets with the subgraphs. Then we have:

$$\sum_j w(F_{i,j}) \leq w(F) + \frac{2}{\ell} \cdot w(F)$$

for some specific integer i .

The interior vertices of each subgraph have their weights counted only once towards the sum $\sum_j w(F_{i,j})$.

The same does not hold for the vertices on the boundary, they are counted once in $F_{i,j}$ and once in $F_{i,j+1}$. To account for that we rewrite the sum on the LHS as:

1. $\sum_j w(F_{i,j}) = w(F) + w(\{\text{vertices on the boundary}\})$.

2. Now we can rewrite the RHS of 1 as:

$$w(F) + w(\{\text{vertices on the boundary}\}) = w(F) + w(OPT_i) + w(OPT_{i+1}) \text{ for some } i \in \{1, 2, \dots, \ell - 1\}.$$

3. From Lemma 4.7 we have an integer $i^* \in \{1, 2, \dots, \ell - 1\}$, such that:

$$w(OPT_{i^*}) + w(OPT_{i^*+1}) \leq \frac{2}{\ell} \cdot w(F).$$

4. For the integer i^* in step 3 we substitute in step 2 and get:

$$w(F) + w(OPT_{i^*}) + w(OPT_{i^*+1}) \leq w(F) + \frac{2}{\ell} \cdot w(F).$$

Finally, we chain the inequalities we obtained in Lemma 4.5, Result 4.6, and Lemma 4.8 to argue that for some $i^* \in \{1, 2, \dots, \ell - 1\}$:

$$w(S_{i^*}) \leq \sum_j w(S_{i^*,j}) \leq \sum_j w(F_{i^*,j}) \leq w(F) + \frac{2}{\ell} \cdot w(F)$$

Algorithm 6 Exact algorithm for weighted FVS-BCL for each subgraph $G_{i,j}$

Require: ℓ -outerplanar graph $G = (V, E)$, $w : V \rightarrow \mathbb{R}$

Ensure: Exact solution S

1: **return** S

THINGS TO PROVE/CLARIFY:

1. justification for outerplanarity (easy?) for each subgraph, i.e. $G_{i,j}$ is ℓ -outerplanar (because of BFS).
2. feasibility of the union solution, prove that S_i is indeed a solution, here make use of the number of overlap layers (medium)
3. clarify how we use the shift and slice variables, what does $\forall(i, j)$ mean, explain we actually talk about the subgraphs, what does i^* mean.
4. for lemma 4.7 and lemma 4.8,

4.3 Baker's technique for weighted FVS-BCL of arbitrary cycle length

In this subsection we generalize what we saw in subsection 4.2 and inspect what happens when we try to break all cycles of length ρ . To reiterate, our algorithm is only interested in breaking cycles of length ρ .

Algorithm 7 Baker's technique for the weighted FVS-BCL

Require: Planar graph $G = (V, E)$, parameter $\ell \leftarrow \frac{1}{\epsilon}$, $w : V \rightarrow \mathbb{R}$, ρ - integer specifying cycles of what length to break

Ensure: $(1 + \epsilon)$ -approximation for FVS-BCL

- 1: Perform BFS from some arbitrary vertex r
 - 2: $S \leftarrow \emptyset$
 i : *shift*; j : *slice*
 - 3: **for** each $i = 0$ to $\ell - 1$ **do**
 - 4: Let $G_{i,j}$ be the subgraph induced on vertices at levels $j \cdot \ell + i$ through $(j + 1) \cdot \ell + i + \lfloor \frac{\rho}{2} \rfloor$ for $0 \leq i < \ell$.
 - 5: $\forall j$ compute $S_{i,j}$, the min. weighted FVS-BCL of $G_{i,j}$ using Algorithm 8 as a subroutine.
 - 6: $\forall j$ let $S_i = \bigcup_j S_{i,j}$
 - 7: $S \leftarrow S \cup S_i$
 - 8: **end for**
 - 9: **return** S_{i^*} from S , such that $w(S_{i^*}) \leq w(S_i), \forall i \in \{1, 2, \dots, \ell - 1\}$.
-

The changes between Algorithm 5 and Algorithm 7 pertain to lines 4-7. The main difference is on line 4 where we define the subgraphs we want to inspect. This change enables us to argue that the union of the

local solutions, defined on line 6, is indeed a feasible solution for the whole graph. What we then have to prove is that this solution is not only feasible, but for some integer i^* , it is optimal.

In Lemma 4.9 we present the two results we got in Lemma 4.5 and Result 4.6 for arbitrary cycle length.

Lemma 4.9: Optimum solution bound for FVS- ρ -BCL

Let $G = (V, E)$ be a planar graph, $w : V \rightarrow \mathbb{R}$ be a weight function and ρ an integer. Let $F \equiv OPT$ be the optimum solution in the whole graph for FVS- ρ -BCL and $F_{i,j} = G_{i,j} \cap F$. Let $S_{i,j}$ be an optimum solution for $G_{i,j}$ for some shift i and slice j . We claim that:

$$w(S_i) \leq \sum_j w(S_{i,j}) \leq \sum_j w(F_{i,j}).$$

We first prove the second inequality by choosing random shift and slice variables and seeing that the inequality holds. In this step we deviate from Lemma 4.5 in that when we choose (a, b) to be the aforementioned variables, we inspect the induced subgraph $G_{a,b}$ on levels $b \cdot \ell + a$ through $(b + 1) \cdot \ell + a + \lfloor \frac{\ell}{2} \rfloor$. In that subgraph, the optimum solution $S_{a,b}$ found on step 5 of Algorithm 7 is better than the feasible solution $F_{a,b}$ for $G_{a,b}$.

The first inequality holds, because the vertices on the boundaries are counted twice towards the RHS of the inequality $w(S_i) \leq \sum_j w(S_{i,j})$.

Algorithm 8 Exact algorithm for weighted FVS-BCL of cycle length ρ

Require: ℓ -outerplanar graph $G = (V, E)$, $w : V \rightarrow \mathbb{R}$, ρ - cycle length

Ensure: Exact solution S

1: **return** S

4.4 Exact algorithm for FVS-BCL

In this subsection we are looking for an algorithm that finds exact solutions to the FVS-BCL problem in each subgraph. For each subgraph we can find the treewidth.

We take a logical approach that was introduced by Courcelle [23]. He shows that graph properties that can be stated in monadic second-order logic (MSOL) can be solved in linear time for graphs of bounded treewidth. In the following two subsections we briefly explain MSOL and then apply it to our problem. We follow a survey on the matter in Satzinger [24].

4.4.1 MSOL - a language for graph problems

We want to introduce a language that translates graph problems to formulas, therefore reducing the problem to a satisfiability problem. Consequently, our goal is to find a formula Φ with the property:

The problem Π is solvable for a graph G iff $G \models \Phi$, i. e. the formula Φ is satisfied by the graph G .

Problem Π is solvable in polynomial (in this case linear) time for graphs of treewidth at most k .

Monadic second-order logic is an extension of first-order logic. Additionally to having object variables $x, y, \dots$, logical connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ and quantifiers for object variables $\forall x, \exists x$, MSOL considers sets of object variables $X, Y, \dots$, and relation to them via $\in$.

Whenever graph problems are involved, typically we express solution through vertices and edges. Additionally, as we mentioned, we can construct sets of vertices and edges. Therefore we have the following machinery:

- Object variables: *vertices*: $v_1, v_2, \dots$ and *edges*: $e_1, e_2, \dots$.
- Set variables: sets of vertices $V_1, V_2, \dots$ and sets of edges $E_1, E_2, \dots$.
- Binary relation $\in$: $\{\text{object variable}\} \times \{\text{set variable}\} \rightarrow \{0, 1\}$. Therefore $v \in V$ iff v is an element of the corresponding set V .
- The $Adj(e, v_i, v_j)$ relation. It detects whether edge e is an edge from vertex v_i to vertex v_j where $v_i \neq v_j$.

With this machinery, one obtains a first-order language suitable for expressing relations in a graph. To extend this to a traditional second-order logic language, one must add quantifications over set variables.

- Quantification over set variables: $\forall V_i, \forall E_i$ and $\exists V_i, \exists E_i$.

4.4.2 Exact algorithm for FVS-BCL using MSOL

Courcelle's work was extended to extended monadic second-order logic [25, 26, 27]. These extensions allow to optimize over the size and weight of a free set variable.

The minimum weight FVS-BCL for a length $\rho = 4$ can be expressed as:

$$\begin{aligned}
& \min_{F \subseteq V} w(F) : \forall v \forall u \forall w \forall z : \\
& \quad v \in V \wedge u \in V \wedge w \in V \wedge z \in V \\
& \quad \wedge v \neq w \wedge u \neq z \\
& \quad \wedge (v, u) \in E \wedge (u, w) \in E \wedge (w, z) \in E \wedge (v, z) \in E \\
& \quad \rightarrow v \in F \vee u \in F \vee w \in F \vee z \in F
\end{aligned}$$

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